

Prerna Kumari

Software Engineer II | Backend & Distributed Systems

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Professional Summary

Software Engineer II with **3 years** of experience leading development of distributed streaming microservices and data workflows in cloud-native environments using **C#/.NET Core Kafka** event pipelines, **REST APIs**, and container deployments on **Kubernetes** and **OpenShift**. Strong **system design and architecture** capabilities; proven track record of leading moderately complex initiatives, collaborating with cross-functional teams, and providing technical guidance to less experienced engineers.

Experience

S&P Global

Software Engineer II

Gurugram, IN

Oct 2025 – Present

- Led design and development of a **REST API microservice** using **ASP.NET Core (.NET 10)** to serve real-time financial datasets, supporting JSON and Protobuf; implemented **Swagger/OpenAPI** and async REST endpoints; acted as **technical escalation point** for API design decisions across cross-functional teams.
- Designed high-throughput **Kafka** event streaming pipelines (producer/consumer groups, partitioned topics, offset management) for financial data ingestion into **Apache Cassandra/SQL**; and a **Kafka-to-GCP Pub/Sub** bridge processing **250+ asset classes** at scale to cloud platform(BigQuery).
- Containerized services using **Docker** and deployed to **Amazon EKS** via **Azure DevOps CI/CD** pipelines, with secrets managed via **AWS Secrets Manager**, and removed the dependency on Windows Server.
- Partner with product teams developing **Model Context Protocol (MCP)** capabilities that integrate enterprise financial data into AI-powered experiences including **ChatIQ in CIQ Pro**. Improve developer experience by refining tool schemas, parameter definitions, documentation, and AI interaction patterns to produce more reliable and intuitive model behavior.
- Work extensively with **GitHub Copilot** and AI-assisted development workflows, evaluating how AI capabilities can improve engineering productivity while maintaining software quality and security standards.
- Collaborated with peers and mid-level engineers to lead **system design reviews** and resolve moderately complex technical challenges; provided guidance and direction to less experienced engineers; leveraged **GitHub Copilot** to accelerate delivery and raise code review quality.

Software Engineer I

Dec 2023 – Sep 2025

- Developed containerized **.NET C#** microservices for Fixed Income data ingestion from **AWS S3**, deployed as **Kubernetes CronJobs** across 10+ repos; modernized pipelines (NTLM to OAuth, Secrets Manager, .NET 10 upgrade) **reducing failures by 95%**.

Signant Health (DSG Inc)

Software Engineer I

Noida, IN

Sep 2022 – Nov 2023

- Optimized complex **SQL** queries and data access patterns, **improving reporting pipeline performance by 30%**.

Defence Research & Development Organisation (DRDO)

Trainee

Delhi, IN

Jan 2022 – Jun 2022

Skills

Programming & Frameworks: C#, .NET Core / .NET Framework (.NET 6–10), Python, REST API, Swagger/OpenAPI, Microservices, Distributed Streaming, Event-Driven Architecture, Distributed Systems, Clean Architecture, OOP

Databases & Messaging: PostgreSQL, SQL Server (MSSQL), Apache Cassandra, Redis, Apache Kafka (Producer/Consumer), Elasticsearch

Cloud & DevOps: AWS (EKS, Secrets Manager), GCP (Pub/Sub), BigQuery, Azure DevOps, Jenkins, Docker, Kubernetes, OpenShift, CI/CD Pipelines, Git, Splunk, OpenSearch, Prometheus, Jira, Monitoring & Observability

AI / ML & Practices: LLMs, Prompt Engineering, RAG, GitHub Copilot, Performance Optimization, Agile/Scrum

Education

Dr. A.P.J. Abdul Kalam Technical University

B.Tech, Computer Science and Engineering, 2023 | 82%

Central Board of Secondary Education

Class XII, 2019 | 73.4% Class X, 2017 | CGPA 9.1

Projects

AI Resume Matcher

github.com/prernak1/AI-Resume-Matcher

ASP.NET Core 9 · C# · Clean Architecture · Greenhouse API · Vanilla JS

- Built an **ASP.NET Core 9 Web API** backend with NLP tokenization-based resume parsing to extract candidate skill tokens; integrated the **Greenhouse Harvest API** using **HttpClient + Task.WhenAll** for concurrent live job aggregation from multiple companies (Stripe, Airbnb, Shopify etc.).
- Implemented a job-relevance scoring algorithm using **TF-IDF-style term weighting and cosine similarity** — ranks openings by skill-overlap score between parsed resume tokens and required/preferred job skill sets.